19-206-0711 ENERGY MANAGEMENT AND CONSERVATION

Module – 1

Energy Scenario

Course Outcomes

- On completion of this course the student will be able to:
  1. Calculate the efficiency of various thermal utilities.
  2. Design suitable energy monitoring system to analyze and optimize the energy consumption in an organization.
  3. Design suitable systems for heat recovery and co-generation to improve thermal efficiency.
  4. Relate energy audit methods learnt to identify the areas deserving tighter control to save energy expenditure.
  5. Explain to the employees of the organization about the need and the methods of energy conservation

Syllabus

- Module I
  - Energy Scenario
  - Basics of Energy its various forms and conservation
- Module II
  - Evaluation of thermal performance
  - Energy Management & Audit
- Module III
  - Energy Monitoring and Targeting
- Module IV
  - Energy Efficiency in Thermal Utilities and systems
  - Heat Recovery and Co-generation

References

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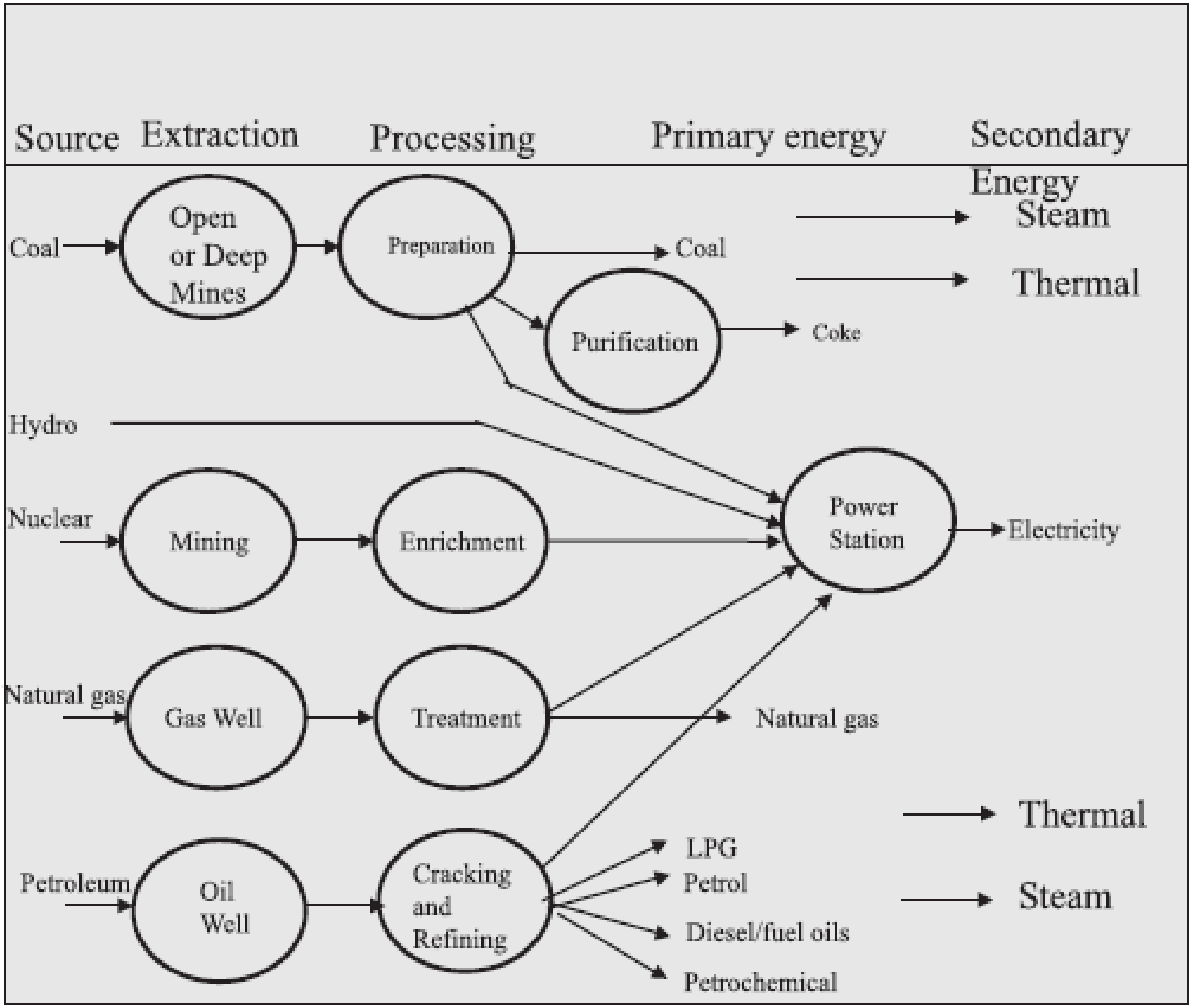
Introduction

- Energy is one of the major inputs for the economic development of any country.
- In the case of the developing countries, the energy sector assumes a critical importance in view of the ever-increasing energy needs requiring huge investments to meet them.
- Energy can be classified into several types based on the following criteria:
  - Primary and Secondary energy
  - Commercial and Non commercial energy
  - Renewable and Non-Renewable energy

Primary and Secondary Energy

- Primary energy sources are those that are either found or stored in nature.
- Common primary energy sources are coal, oil, natural gas, and biomass (such as wood).
- Other primary energy sources available include nuclear energy from radioactive substances, thermal energy stored in earth's interior, and potential energy due to earth's gravity.
- Primary energy sources are mostly converted in industrial utilities into secondary energy sources; for example coal, oil or gas converted into steam and electricity.
- Primary energy can also be used directly.
- Some energy sources have non-energy uses, for example coal or natural gas can be used as a feedstock in fertiliser plants.

Major Primary and Secondary Energy Sources



Commercial Energy

- The energy sources that are available in the market for a definite price are known as commercial energy.
- By far the most important forms of commercial energy are electricity, coal and refined petroleum products.
- Commercial energy forms the basis of industrial, agricultural, transport and commercial development in the modern world.
- In the industrialized countries, commercialized fuels are predominant source not only for economic production, but also for many household tasks of general population.
- Examples:
  - Electricity
  - Lignite
  - Coal
  - Oil
  - Natural gas

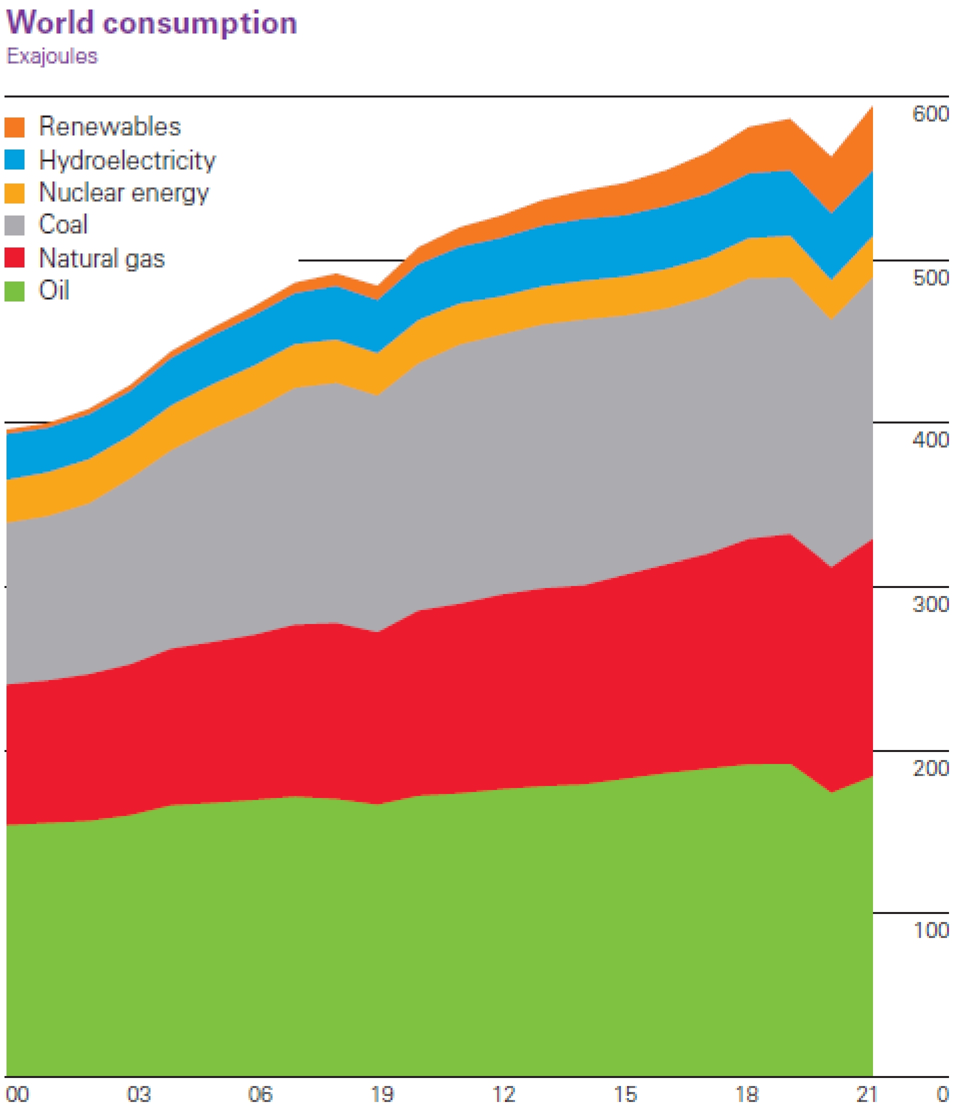
Non-Commercial Energy

- The energy sources that are not available in the commercial market for a definite price are classified as non-commercial energy.
- Non-commercial energy sources include fuels such as firewood, cattle dung and agricultural wastes, which are traditionally gathered, and not bought at a price used especially in rural households.
- These are also called traditional fuels.
- Non-commercial energy is often ignored in energy accounting.
- Examples:
  - Firewood
  - Agro waste in rural areas
  - Solar energy for
    - water heating
    - electricity generation
    - drying grain, fish and fruits
  - Animal power for transport, threshing, lifting water for irrigation, crushing sugarcane
  - Wind energy for lifting water and electricity generation

Renewable and Non-Renewable Energy

- Renewable energy
  - Energy obtained from sources that are essentially inexhaustible.
  - Examples
    - Wind power
    - Solar power
    - Geothermal energy
    - Tidal power
    - Hydroelectric power
  - The most important feature of renewable energy is that it can be harnessed without the release of harmful pollutants.
- Non-renewable energy
  - Energy obtained from sources that are likely to deplete with time
  - Conventional fossil fuels such as coal, oil and gas.

Global Energy Consumption



Indian Energy Scenario

- Coal is the country's top energy source with a share of 44% in 2021, followed by oil (24%) and biomass (22%).
- Natural gas covers 6% and primary electricity (hydro, nuclear, solar, and wind) 4%.
- Total energy consumption per capita is around 25.4 GJ (2021).
- Total energy consumption reached 927 Mtoe in 2021.

Coal Supply in India

- India has huge coal reserves, at least 84,396 million tonnes of proven recoverable reserves (at the end of 2003).
- This amounts to almost 8.6% of the world reserves and it may last for about 230 years at the current Reserve to Production (R/P) ratio.
- In contrast, the world's proven coal reserves are expected to last only for 192 years at the current R/P ratio.
- Reserves/Production (R/P) ratio- If the reserves remaining at the end of the year are divided by the production in that year, the result is the length of time that the remaining reserves would last if production were to continue at that level.
- India is the fourth largest producer of coal and lignite in the world.
- Coal production is concentrated in these states of Andhra Pradesh, Uttar Pradesh, Bihar, Madhya Pradesh, Maharashtra, Orissa, Jharkhand, and West Bengal.

Indian Energy Scenario

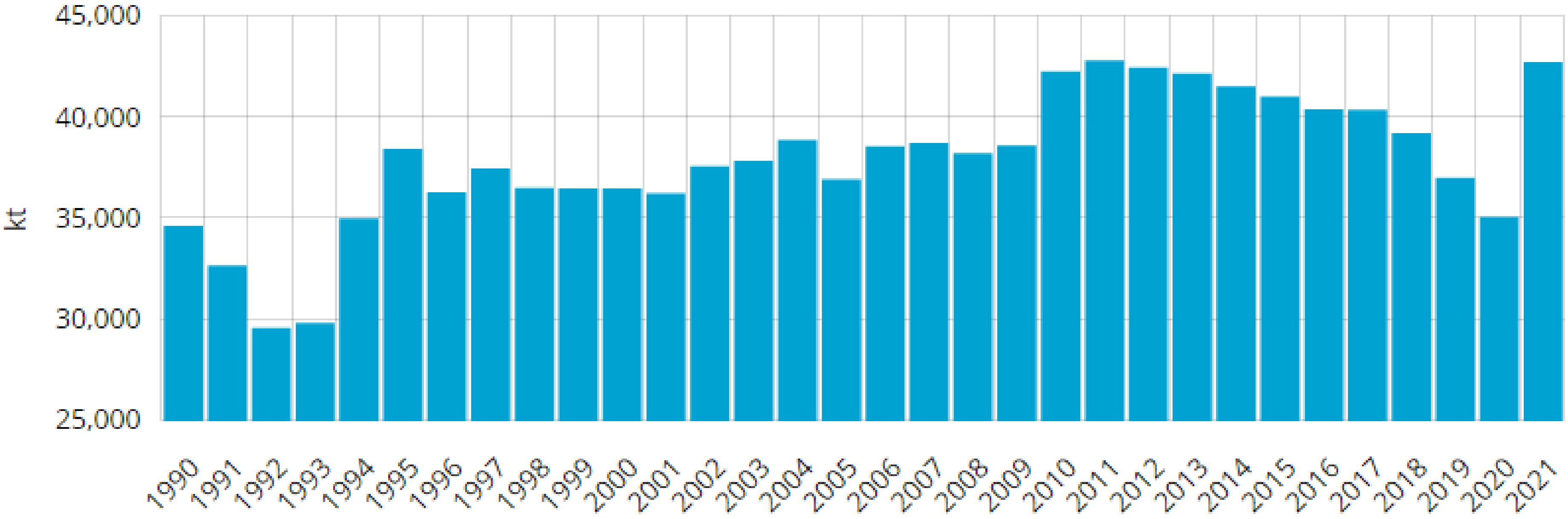
Year	Crude Oil Production (MMT)	% Growth in Crude Oil Production	Natural Gas Production (BCM)	% Growth in Natural Gas Production
1960-61	0.45	-	-	-
1970-71	6.82	1.49	1.45	5.07
1980-81	10.51	-10.71	2.40	-13.36
1990-91	33.02	-3.14	18.00	5.88
2000-01	32.45	1.56	29.48	3.79
2010-11	37.71	11.94	52.22	9.95
2011-12	38.09	1.00	47.56	-8.93
2012-13	37.86	-0.60	40.68	-14.47
2013-14	37.79	-0.19	35.41	-12.96
2014-15	37.46	-0.87	33.66	-4.94
2015-16	36.94	-1.39	32.25	-4.18
2016-17	36.01	-2.53	31.90	-1.09
2017-18	35.68	-0.90	32.65	2.36
2018-19	34.20	-4.15	32.87	0.69
2019-20	32.17	-5.95	31.18	-5.14
2020-21(P)	**30.49**	**-5.21**	**28.67**	**-8.05**
CAGR 2011-12 to 2020-21 (%)	**-2.4**		**-5.5**	

P : Provisional

Oil Supply in India

- Oil production exceeded to 43 Mt in 2021.
- India is the world's third largest importer of crude oil, with imports of 210 Mt in 2021.
- In 2021, crude oil imports represented 81% of the country's oil needs.
- The country is a net oil product exporter (24 Mt in 2021), although its exports have fallen by 29% since 2015.

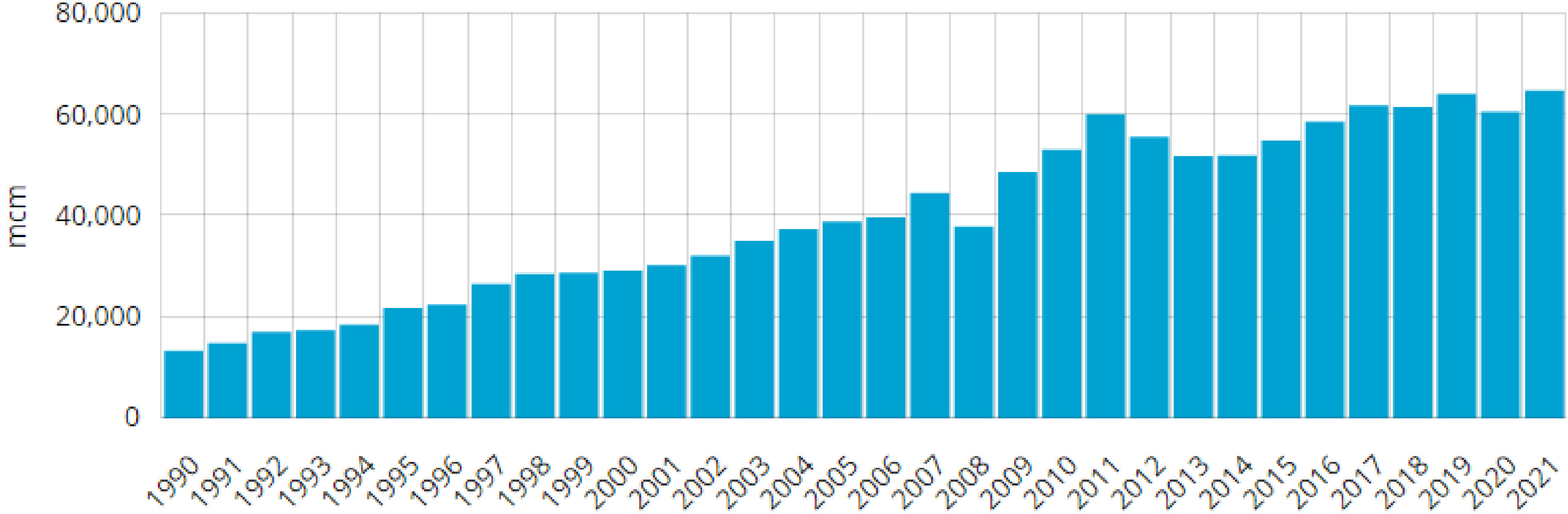
Oil Supply in India



Natural Gas Supply in India

- Natural gas consumption reached 65 bcm in 2021
- Industry is the main consumer of gas with 48% in 2021 (mainly fertiliser plants). It is followed by electricity production (25%) and, to a much lesser extent, transport (5%) and the residential and services sector (7%).
- India aims to promote the use of CNG vehicles to reduce pollution in cities.
- 3,628 CNG stations have been deployed as of December 2021.

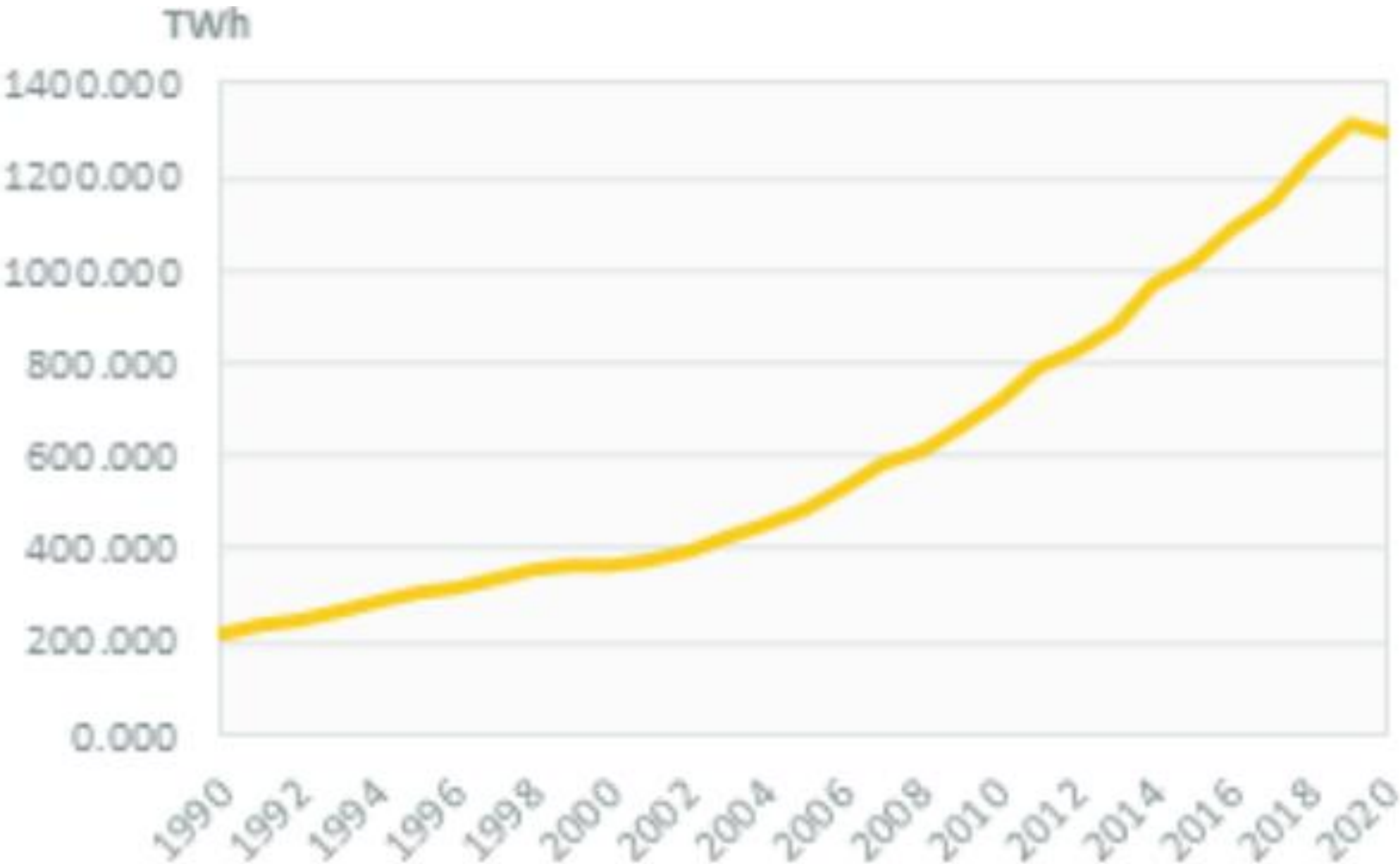
Natural Gas Supply in India



Electrical Energy Supply in India

- Electricity consumption increased by around 5% in 2021 to 1355 TWh.
- It grew rapidly over 2010-2019 (7%/year).

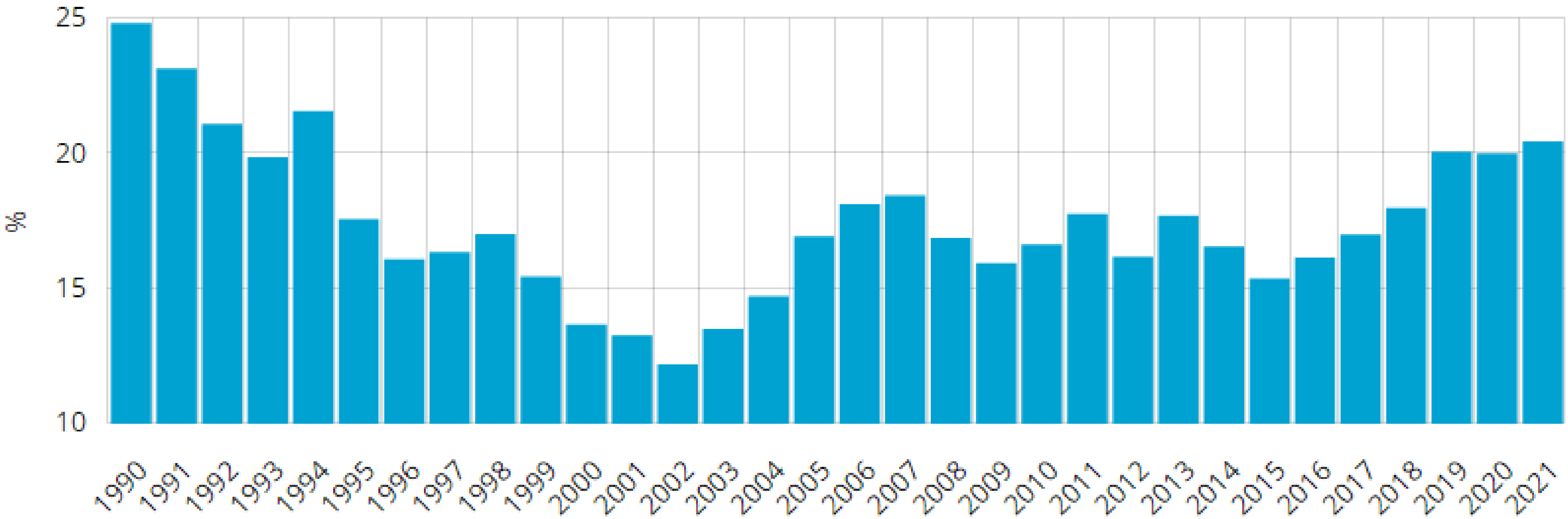
Electrical Energy Supply in India



Electrical Energy from Renewable Sources in India

- IREDA, the Indian Renewable Energy Development Agency, finances the development of renewables.
- SECI, Solar Energy Corporation of India, under MNRE, manages the implementation of the Jawaharlal Nehru National Solar Mission (JNNSM), which aims to deploy 100 GW of solar power over 2010-2022 (mid-2022, only half of this target is achieved).

Electrical Energy from Renewable Sources in India



Final Energy Consumption

- Final energy consumption is the actual energy demand at the user end.
- This is the difference between primary energy consumption and the losses that takes place in transport, transmission & distribution and refinement.

TABLE 1.2 DEMAND FOR COMMERCIAL ENERGY FOR FINAL CONSUMPTION (BAU SCENARIO)

Source	Units	1994-95	2001-02	2006-07	2011-12
Electricity	Billion Units	289.36	480.08	712.67	1067.88
Coal	Million Tonnes	76.67	109.01	134.99	173.47
Lignite	Million Tonnes	4.85	11.69	16.02	19.70
Natural Gas	Million Cubic Meters	9880	15730	18291	20853
Oil Products	Million Tonnes	63.55	99.89	139.95	196.47

Source: Planning Commission *BAU:_Business As Usual*

Sector wise Energy Consumption in India

- The major commercial energy consuming sectors in the country are classified as shown in the Figure.
- As seen from the figure, industry remains the biggest consumer of commercial energy and its share in the overall consumption is 49%. (Reference year: 1999/2000)

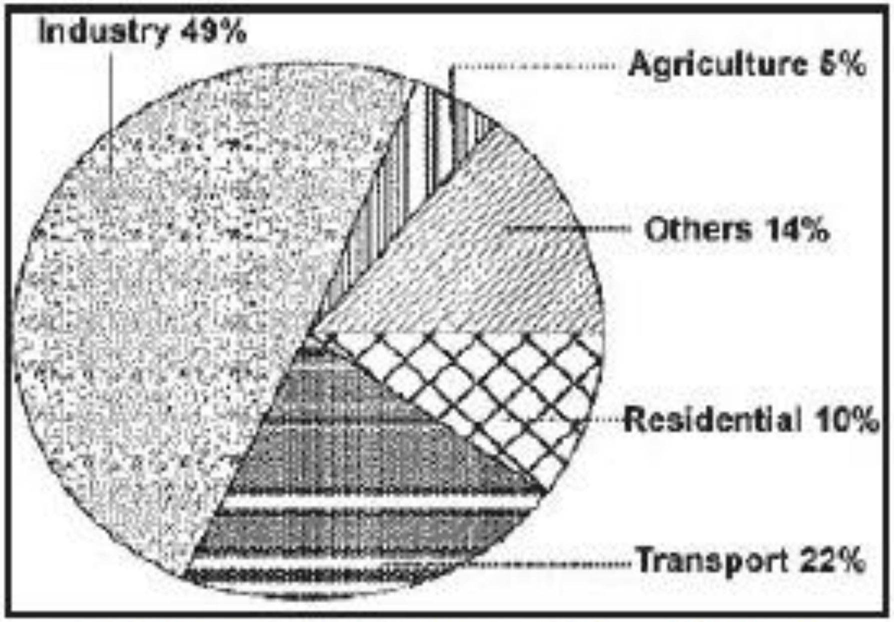


Figure 1.5 Sector Wise Energy Consumption (1999-2000)

Energy Needs of Growing Economy

- Economic growth is desirable for developing countries, and energy is essential for economic growth.
- However, the relationship between economic growth and increased energy demand is not always a straightforward linear one. For example, under present conditions, 6% increase in India's Gross Domestic Product (GDP) would impose an increased demand of 9 % on its energy sector.
- The ratio of energy demand to GDP is a useful indicator.
- A high ratio reflects energy dependence and a strong influence of energy on GDP growth.
- The developed countries, by focusing on energy efficiency and lower energy-intensive routes, maintain their energy to GDP ratios at values of less than 1.
- The ratios for developing countries are much higher.

Energy Needs of Growing Economy

- Per Capita Energy Consumption
  - The per capita energy consumption is too low for India (25.4 GJ in 2021) as compared to developed countries.
  - The world average per capita energy consumption is 75.6.
  - The per capita consumption is likely to grow in India with growth in economy thus increasing the energy demand.
- Energy intensity
  - Energy intensity is energy consumption per unit of GDP.
  - Energy intensity indicates the development stage of the country.
  - India's energy intensity is 3.7 times of Japan, 1.55 times of USA, 1.47 times of Asia and 1.5 times of World average.

Long Term Energy Scenario for India

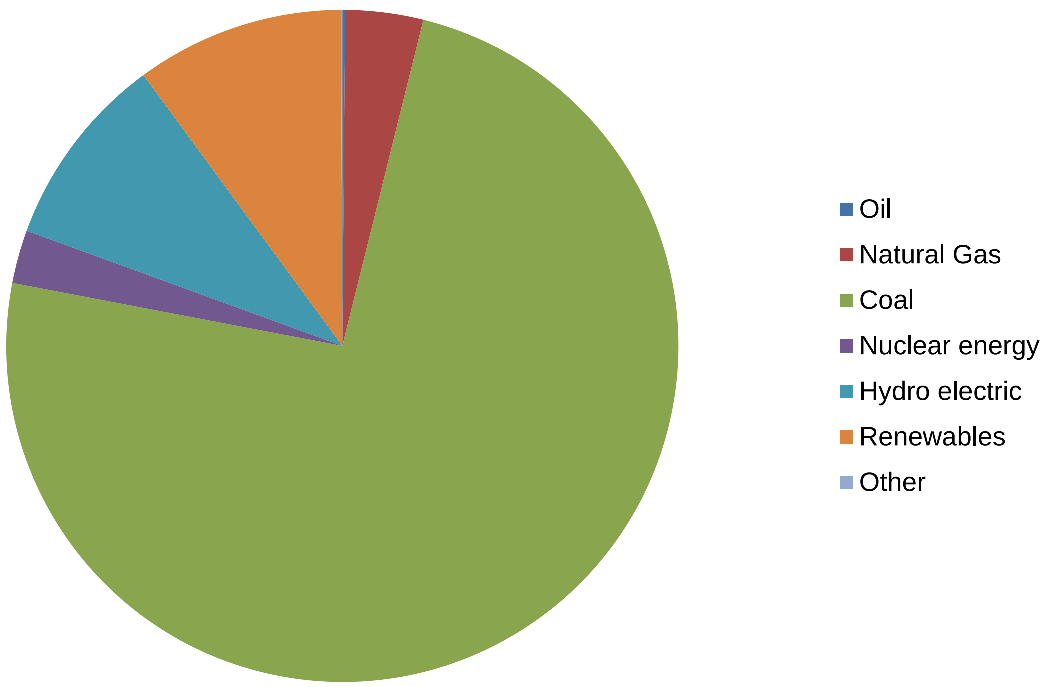
- Coal
  - Coal is the predominant energy source for power production in India.
  - Despite significant increases in total installed capacity during the last decade, the gap between electricity supply and demand continues to increase.
  - The resulting shortfall has had a negative impact on industrial output and economic growth.
  - However, to meet expected future demand, indigenous coal production will have to be greatly expanded.
  - Indian coal is typically of poor quality and as such requires to be beneficiated to improve the quality
  - Coal imports will also need to increase dramatically to satisfy industrial and power generation requirements.

Long Term Energy Scenario for India

- Oil
  - Oil production in India decreased from 937,000 barrels per day in 2011 to 746,000 barrels per day in 2021.
  - Oil consumption in India increased from 3,475,000 barrels per day in 2011 to 4,878,000 barrels per day in 2021.
  - Most of India's total oil demand may have to be met by imports in future.
- Natural gas
  - India's natural gas production has fallen from 42.9 billion cubic meters in 2011 to 28.5 billion cubic meters in 2021.
  - The natural gas consumption has risen from 60.3 billion cubic meters in 2011 to 62.2 billion cubic meters in 2021.

Long Term Energy Scenario for India

- Electricity
  - Electricity generation in India increased from 1034 TWh in 2011 to 1714.8 TWh in 2021.
  - India's peak power deficit decreased from 16.6% in 2008 to 0.4% in 2021.
  - India still has electricity demand shortage.



Energy Pricing in India

- Price of energy does not reflect true cost to society.
- The basic assumption underlying efficiency of market place does not hold in our economy, since energy prices are undervalued and energy wastages are not taken seriously.
- Pricing practices in India like many other developing countries are influenced by political, social and economic compulsions at the state and central level.
- More often than not, this has been the foundation for energy sector policies in India.
- The Indian energy sector offers many examples of cross subsidies e.g., diesel, LPG and kerosene being subsidised by petrol, petroleum products for industrial usage and industrial, annd commercial consumers of electricity subsidising the agricultural and domestic consumers.

Energy Pricing in India

- Coal
  - Grade wise basic price of coal at the pithead excluding statutory levies for run-of-mine (ROM) coal are fixed by Coal India Ltd from time to time.
  - The pithead price of coal in India compares favourably with price of imported coal.
  - In spite of this, industries still import coal due its higher calorific value and low ash content.
- Oil
  - As part of the energy sector reforms, the government has attempted to bring prices for many of the petroleum products (naphtha, furnace oil, LSHS, LDO and bitumen) in line with international prices.
  - The most important achievement has been the linking of diesel prices to international prices and a reduction in subsidy.
  - LPG, consumed mainly by domestic sectors, was heavily subsidised till June 2020.

Energy Pricing in India

- Natural Gas
  - The government has been the sole authority for fixing the price of natural gas in the country.
  - It has also been taking decisions on the allocation of gas to various competing consumers.
  - The CNG price varies from Rs 75 to Rs 90 per kg.
- Electricity
  - Electricity tariffs in India are structured in a relatively simple manner.
  - The price per kWh varies significantly across States as well as customer segments within a State.
  - Tariffs in India have been modified to consider the time of usage and voltage level of supply.
  - In addition to the base tariffs, some State Electricity Boards have additional recovery from customers in form of fuel surcharges, electricity duties and taxes.

Energy Security

- The basic aim of energy security for a nation is to reduce its dependency on the imported energy sources for its economic growth.
- India will continue to experience an energy supply shortfall throughout the forecast period.
- This gap has widened since 1985, when the country became a net importer of coal.
- India has been unable to raise its oil production substantially in the 1990s.
- Rising oil demand of close to 10 percent per year has led to sizable oil import bills.

Energy Security

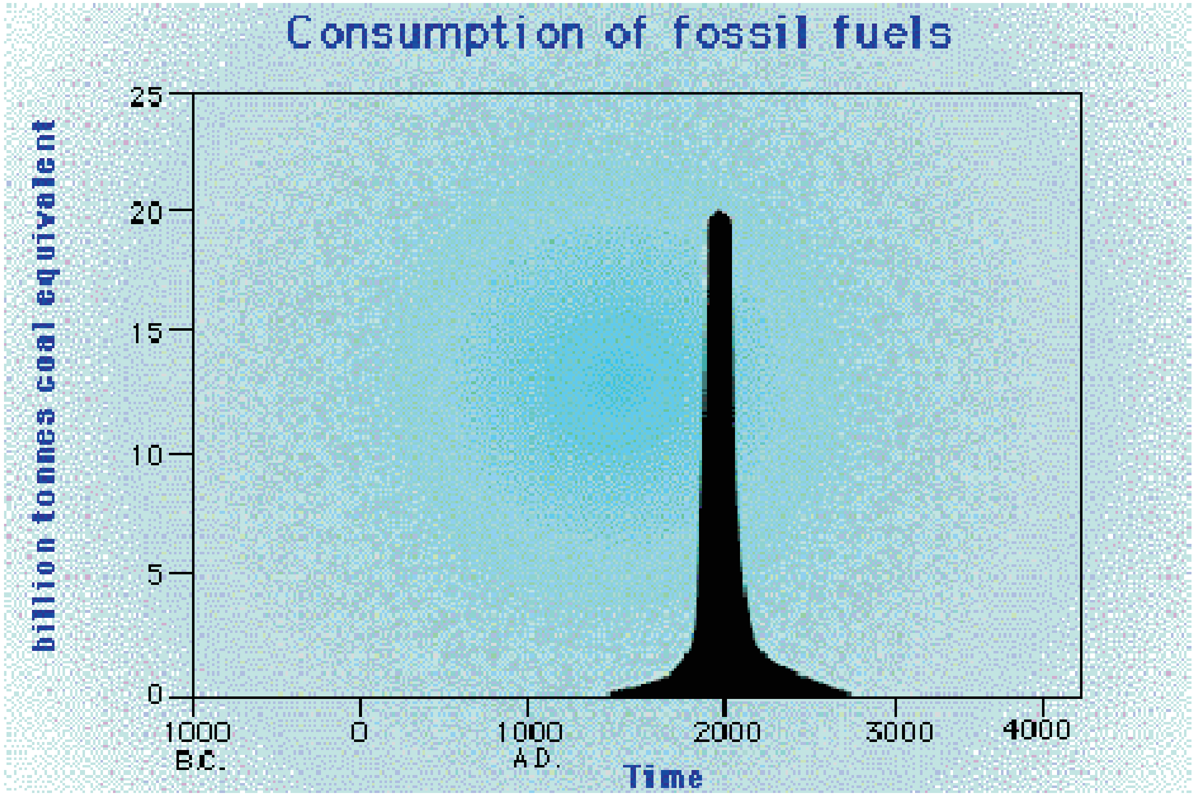
- Imports of oil and coal have been increasing at rates of 3.4% and 7.5% per annum respectively during the period 2011-2021.
- The dependence on energy imports is projected to increase in the future.
- The imports of gas and LNG (liquefied natural gas) are likely to increase in the coming years.
- This energy import dependence implies vulnerability to external price shocks and supply fluctuations, which threaten the energy security of the country.
- Increasing dependence on oil imports means reliance on imports from the Middle East, a region susceptible to disturbances and consequent disruptions of oil supplies.
- This calls for diversification of sources of oil imports.
- The need to deal with oil price fluctuations also necessitates measures to be taken to reduce the oil dependence of the economy, possibly through fiscal measures to reduce demand, and by developing alternatives to oil, such as natural gas and renewable energy.

Energy Security

- Some of the strategies that can be used to meet future challenges to their energy security are
  - Building stockpiles
  - Diversification of energy supply sources
  - Increased capacity of fuel switching
  - Demand restraint
  - Development of renewable energy sources
  - Energy efficiency
  - Sustainable development
- The simplest and the most easily attainable is reducing demand through persistent energy conservation efforts.

Energy Conservation and its Importance

- Coal and other fossil fuels, which have taken three million years to form, are likely to deplete soon.
- In the last two hundred years, we have consumed 60% of all resources.
- For sustainable development, we need to adopt energy efficiency measures.
- The nonrenewable reserves are continually diminishing with increasing consumption and will not exist for future generations.

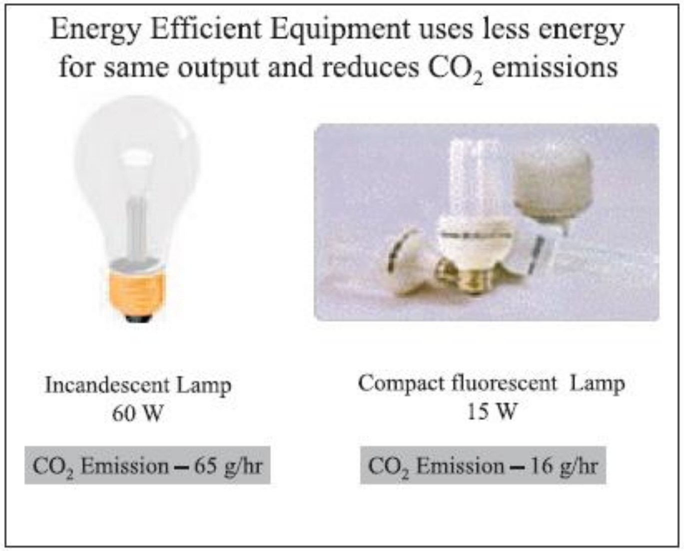


What is Energy Conservation?

- Energy conservation and energy efficiency are separate, but related concepts.
- Energy conservation is achieved when growth of energy consumption is reduced, measured in physical terms.
- Energy conservation can, therefore, be the result of several processes or developments, such as productivity increase or technological progress.

Energy Efficiency for Energy Conservation

- Energy efficiency is often viewed as a resource option like coal, oil or natural gas.
- It provides additional economic value by preserving the resource base and reducing pollution. For example, replacing traditional light bulbs with Compact Fluorescent Lamps (CFLs) means you will use only 1/4th of the energy to light a room. Pollution levels also reduce by the same amount
- Energy efficiency is achieved when energy intensity in a specific product, process or area of production or consumption is reduced without affecting output, consumption or comfort levels.
- Promotion of energy efficiency will contribute to energy conservation and is therefore an integral part of energy conservation promotional policies.
- Very simply, energy efficiency means using less energy to perform the same function.



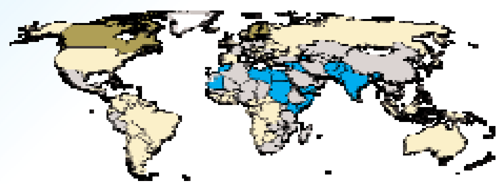
Energy Efficiency Benefits

Energy Efficiency Benefits

Industry

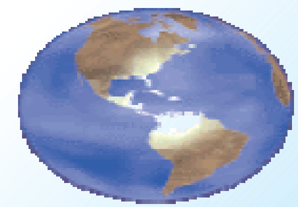
- Reduced energy bills
- Increased Competitiveness
- Increased productivity
- Improved quality
- Increased profits !

Nation



- Reduced energy imports
- Avoided costs can be used for poverty reduction
- Conservation of limited resources
- Improved energy security

Globe



- Reduced GHG and other emissions
- Maintains a sustainable environment

Energy Strategy for the Future

- Strategies
  - Immediate strategy
  - medium-term strategy strategy
  - long-term strategy
- Immediate-term strategy:
  - Rationalizing the tariff structure of various energy products.
  - Optimum utilization of existing assets.
  - Efficiency in production systems and reduction in distribution losses, including those in traditional energy sources.
  - Promoting R&D, transfer and use of technologies and practices for environmentally sound energy systems, including new and renewable energy sources.

Energy Strategy for the Future

- Medium-term strategy:
  - Demand management through greater conservation of energy, optimum fuel mix, structural changes in the economy, an appropriate model mix in the transport sector, i.e. greater dependence on rail than on road for the movement of goods and passengers and a shift away from private modes to public modes for passenger transport; changes in design of different products to reduce the material intensity of those products, recycling, etc.
  - There is need to shift to less energy-intensive modes of transport. This would include measures to improve the transport infrastructure viz. roads, better design of vehicles, use of compressed natural gas (CNG) and synthetic fuel, etc. Similarly, better urban planning would also reduce the demand for energy use in the transport sector.
  - There is need to move away from non-renewable to renewable energy sources viz. solar, wind, biomass energy, etc.

Energy Strategy for the Future

- Long-term strategy:
  - Efficient generation of energy resources
    - Efficient production of coal, oil and natural gas
    - Reduction of natural gas flaring
  - Improving energy infrastructure
    - Building new refineries
    - Creation of urban gas transmission and distribution network
    - Maximizing efficiency of rail transport of coal production
    - Building new coal and gas fired power stations.
  - Enhancing energy efficiency
    - Improving energy efficiency in accordance with national, socio-economic, and environmental priorities
    - Promoting of energy efficiency and emission standards

Energy Strategy for the Future

  - Promoting of energy efficiency and emission standards
  - Labeling programmes for products and adoption of energy efficient technologies in large industries
- Deregulation and privatization of energy sector
  - Reducing cross subsidies on oil products and electricity tariffs
  - Decontrolling coal prices and making natural gas prices competitive
  - Privatization of oil, coal and power sectors for improved efficiency.
- Investment legislation to attract foreign investments.
  - Streamlining approval process for attracting private sector participation in power generation, transmission and distribution .

Policy Framework - Energy Conservation Act – 2001

- With the background of high energy saving potential and its benefits, bridging the gap between demand and supply, reducing environmental emissions through energy saving, and to effectively overcome the barrier, the Government of India has enacted the Energy Conservation Act - 2001.
- The Act provides the much-needed legal framework and institutional arrangement for embarking on an energy efficiency drive.
- Under the provisions of the Act, Bureau of Energy Efficiency has been established with effect from 1st March 2002 by merging erstwhile Energy Management Centre of Ministry of Power.
- The Bureau would be responsible for implementation of policy programmes and coordination of implementation of energy conservation activities.

Important Features of The Energy Conservation Act, 2001

- Standards and Labeling
  - Standards and Labeling (S & L) has been identified as a key activity for energy efficiency improvement.
  - The S & L program, when in place would ensure that only energy efficient equipment and appliance would be made available to the consumers.
  - The main provision of EC act on Standards and Labeling are:
    - Evolve minimum energy consumption and performance standards for notified equipment and appliances.
    - Prohibit manufacture, sale and import of such equipment, which does not conform to the standards.
    - Introduce a mandatory labeling scheme for notified equipment appliances to enable consumers to make informed choices
    - Disseminate information on the benefits to consumers

Important Features of The Energy Conservation Act, 2001

- Designated Consumers
  - The main provisions of the EC Act on designated consumers are:
    - The government would notify energy intensive industries and other establishments as designated consumers
    - Schedule to the Act provides list of designated consumers which covered basically energy intensive industries, Railways, Port Trust, Transport Sector, Power Stations, Transmission & Distribution Companies and Commercial buildings or establishments
    - The designated consumer to get an energy audit conducted by an accredited energy auditor
    - Energy managers with prescribed qualification are required to be appointed or designated by the designated consumers
    - Designated consumers would comply with norms and standards of energy consumption as prescribed by the central government.

Important Features of The Energy Conservation Act, 2001

- Certification of Energy Managers and Accreditation of Energy Auditing Firms
  - The main activities in this regard as envisaged in the Act are:
    - A cadre of professionally qualified energy managers and auditors with expertise in policy analysis, project management, financing and implementation of energy efficiency projects would be developed through Certification and Accreditation programme.
    - BEE to design training modules, and conduct a National level examination for certification of energy managers and energy auditors.

Important Features of The Energy Conservation Act, 2001

- Energy Conservation Building Codes:
  - The main provisions of the EC Act on Energy Conservation Building Codes are:
    - The BEE would prepare guidelines for Energy Conservation Building Codes (ECBC)
    - These would be notified to suit local climate conditions or other compelling factors by the respective states for commercial buildings erected after the rules relating to energy conservation building codes have been notified. In addition, these buildings should have a connected load of 500 kW or contract demand of 600 kVA and above and are intended to be used for commercial purposes
    - Energy audit of specific designated commercial building consumers would also be prescribed.

Important Features of The Energy Conservation Act, 2001

- Central Energy Conservation Fund:
  - The EC Act provisions in this case are:
    - The fund would be set up at the centre to develop the delivery mechanism for large-scale adoption of energy efficiency services such as performance contracting and promotion of energy service companies.
    - The fund is expected to give a thrust to R & D and demonstration in order to boost market penetration of efficient equipment and appliances.
    - It would support the creation of facilities for testing and development and to promote consumer awareness.

Important Features of The Energy Conservation Act, 2001

- Bureau of Energy Efficiency (BEE):
  - The mission of Bureau of Energy Efficiency is to institutionalize energy efficiency services, enable delivery mechanisms in the country and provide leadership to energy efficiency in all sectors of economy. The primary objective would be to reduce energy intensity in the Indian Economy.
  - The general superintendence, directions and management of the affairs of the Bureau is vested in the Governing Council with 26 members. The Council is headed by Union Minister of Power and consists of members represented by Secretaries of various line Ministries, the CEOs of technical agencies under the Ministries, members representing equipment and appliance manufacturers, industry, architects, consumers and five power regions representing the states. The Director General of the Bureau shall be the ex-officio member-secretary of the Council.

Important Features of The Energy Conservation Act, 2001

- The BEE will be initially supported by the Central Government by way of grants through budget, it will, however, in a period of 5-7 years become self-sufficient. It would be authorized to collect appropriate fee in discharge of its functions assigned to it. The BEE will also use the Central Energy Conservation Fund and other funds raised from various sources for innovative financing of energy efficiency projects in order to promote energy efficient investment.

- Role of Bureau of Energy Efficiency
  - The role of BEE would be to prepare standards and labels of appliances and equipment, develop a list of designated consumers, specify certification and accreditation procedure, prepare building codes, maintain Central EC fund and undertake promotional activities in co-ordination with center and state level agencies. The role would include development of Energy service companies (ESCOs), transforming the market for energy efficiency and create awareness through measures including clearing house.

Important Features of The Energy Conservation Act, 2001

- Role of Central and State Governments:
  - The following role of Central and State Government is envisaged in the Act
    - Central - to notify rules and regulations under various provisions of the Act, provide initial financial assistance to BEE and EC fund, Coordinate with various State Governments for notification, enforcement, penalties and adjudication.
    - State - to amend energy conservation building codes to suit the regional and local climatic condition, to designate state level agency to coordinate, regulate and enforce provisions of the Act and constitute a State Energy Conservation Fund for promotion of energy efficiency.

Important Features of The Energy Conservation Act, 2001

- Enforcement through Self-Regulation:
  - E.C. Act would require inspection of only two items. The following procedure of self-regulation is proposed to be adopted for verifying areas that require inspection of only two items that require inspection.
  - The certification of energy consumption norms and standards of production process by the Accredited Energy Auditors is a way to enforce effective energy efficiency in Designated Consumers.
  - For energy performance and standards, manufacturer's declared values would be checked in Accredited Laboratories by drawing sample from market. Any manufacturer or consumer or consumer association can challenge the values of the other manufacturer and bring to the notice of BEE. BEE can recognize for challenge testing in disputed cases as a measure for self-regulation.

Important Features of The Energy Conservation Act, 2001

- Penalties and Adjudication:
  - Penalty for each offence under the Act would be in monetary terms i.e. Rs.10,000 for each offence and Rs.1,000 for each day for continued non Compliance.
  - The initial phase of 5 years would be promotional and creating infrastructure for implementation of Act. No penalties would be effective during this phase.
  - The power to adjudicate has been vested with state Electricity Regulatory Commission which shall appoint any one of its member to be an adjudicating officer for holding an enquiry in connection with the penalty imposed.